PABSON Pre-Board Examination -2082

Class: 10	Time: 3:00 hours	F. M.: 75
	Subject: Science & Technology)	

Answer Key:

Section 'A'

1. **Choose the best alternative to the following questions and write in the answer sheet: [10 x 1=10]**

a.	(i) Hyperglycemia
b.	(ii) –CHO
c.	(iv) Digital reputation management
d.	(i) Weber
e.	(iii) 95% O_2 and 5% CO_2
f.	(iii) Boiled tulsi water
g.	(ii) 19.6 m
h.	(iii) 9 N
i.	(ii) kg m^{-1} s^{-2}
j.	(iv) Double displacement reaction

Section 'B'

2. **Write very short answer to the following questions: [9x1=9]**

a. Monocot: Fibrous root system
Dicot: Tap root system

b. **Pascal's Law**
It states that," when a pressure is applied on a liquid enclosed in a vessel then the pressure is transmitted equally in all direction."

c. It is called a Catalyst**.**

d. Froth flotation method

e. Compound used**:** Glycerin (Glycerol)
IUPAC name: Propane-1,2,3-triol

f.

- Soap nut (Reetha)
- Peena

(Other name also can be considered)

g. **DNA test**

h. **Pheromone**

i.

Given:
Heat Energy(Q)=600J
Mass (m) =20 g=0.02 kg
Specific heat capacity (S)=4200 J/kg∘C
We know that ,

Q = msdt

or, 600 = 0.02 x 4200 x dt

or, dt = 600/84

= **7.14°C**

Therefore, rise in temperature = 7.14°C

Section 'C'

Write short answer to the following questions: [14 x2=28]

3. We have,

Electric resistance (ohm, Ω) = kg m^2 s^{-3} A^{-2}

We know,

V = IR [According to Ohm's Law]

or, R = V/I = W/QI = Fxd/ItxI = mxgxd/I^2xt [1]

or, R = = kgxms^{-2}xm/A^2 x s

or, R (Ohm) = kg m^2 s^{-3} A^{-2} [1]

Hence, proved.

4. Any two important characteristics of gymnosperms

1. Seeds are naked [1]

2. They bear cones instead of flowers. [1]

(Other similar points can be considered)

5. Any two differences between worker bee and queen bee. [2]

Worker Bee	Queen Bee
They are sterile female	They are fertile female
They collects nectar and pollen	They lays eggs

(Other similar points can be considered)

6. WBCs protect the body by destroying germs, producing antibodies, and fighting infections by providing immunity power to the body.

[Similar answer can be considered][1]

7. Any two measures to protect rare animals

1. Establish national parks and wildlife reserves. [1]
2. Ban illegal hunting and poaching. [1]

(Other similar points can be considered)

8. (i) Change in gravitational force at point A and B

- Gravitational force is more at point B (distance between them is less to the sun). [1]
- Gravitational force is less at point A (distance between them is more).

(ii) Why does the earth revolve around the sun?
Because of the gravitational force of attraction of the sun [1]

9. P= open universe and R = closed universe [1]

Open Universe	Closed Universe
Expands forever	Expansion stops and contracts

(Other similar points can be considered) [1]

10. Full forms **DDT** → Dichloro-Diphenyl-Trichloroethane [1]

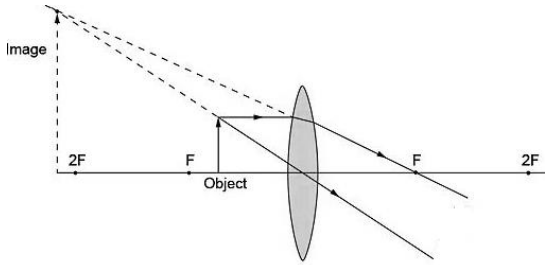
BHC → Benzene Hexa Chloride [1]

11. Any two differences between roasting and calcination [2]

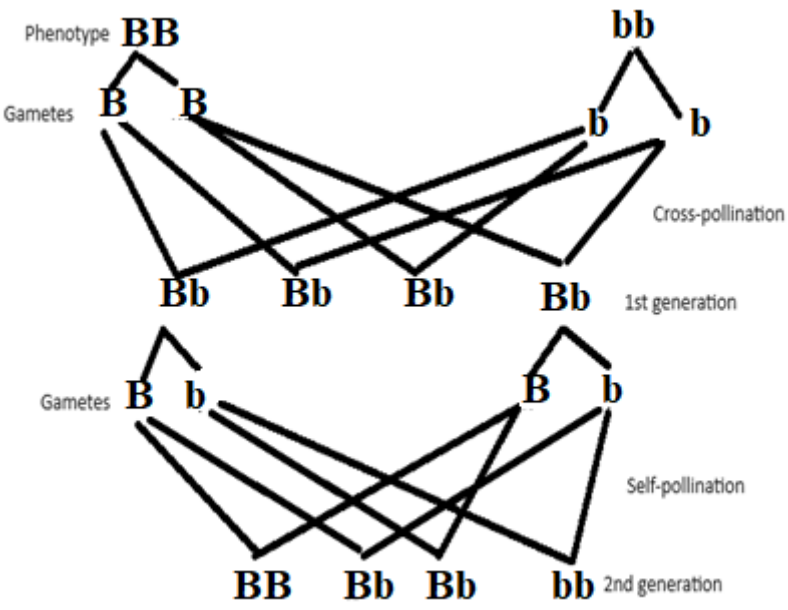
Roasting	Calcination
It is carried in presence of excess air or oxygen	It is carried in absence or limited air/oxygen
It is used for sulphide ores.	It is used for carbonate ores

(Other similar points can be considered)

12. Complete the ray diagram [2]



13. Chart of filial generations (Black × White dog)



14. **Manufacture of ammonia (Haber process)**

Principle

When hydrogen gas and nitrogen gas in the ratio of 3:1 is heated at temperature of about 500°C, under atmospheric pressure 200-900atm. in presence of iron as a catalyst and molybdenum as a promoter, ammonia gas is produced in large scale in industry. [1]

Balanced chemical equation

$3H_2+N_2 \rightleftharpoons 2NH_3$ + Heat energy [1]

15. Burning of magnesium ribbon

(i) Name of substance 'Q' is **Magnesium oxide (MgO)** [1]

(ii) Balanced chemical reaction [1]

$2Mg+O_2$Type equation here.$\rightarrow 2MgO$

16. $F= GMm/R^2$……i

F=mg……ii

From I and ii

$mg= GMm/R^2$ …….[1]

$g= GM/R^2$

$g \alpha \frac{1}{R^2}$ as G and M are constant……[1]

Section 'D'

Write long answer to the following questions [7x4=28]

17. (i) Most reactive metal =B, Most reactive non-metal=X [1]

(ii) Common name is halogen [1]

(iii) $1s^2,2s^22p^6,3s^2$ [1]

(iv) Name of compound = Potassium Chloride [1]

Molecular formula of compound = KCl

18.

(i) Heart attack (or, myocardial infarction) [1]

(ii) Healthy diet and exercise, Avoid harmful habits like smoking, alcohol etc. [Similar points can be considered] [1+1]

(iii) Electrocardiogram (ECG), Angiography ,ECO,Blood test [1]

19.

(i) Refraction of light [1]

(ii) Change in speed of light when it travels from one medium to another medium. [1]

(iii) Incident ray, refracted ray and normal all lies in the same plane. [any others law can be considered] [1]

(iv) μ= Sini/Sinr [1]

20.

(i) Fig-B [1]

(ii) Fig-A, because the number of chromosomes in daughter are equal to the number of chromosomes in mothers cell. So, they are called equational cell division. [1]

(iii) It is (figure 'A') responsible for the growth and development of body. **(Similar points can be considered).** [1]

(iv) Fig-B [1]

21.

(i) Liquid-R has highest specific heat capacity because it has minimum temperature. Since, S α 1/dt [1+1]

(ii) Liquid-P cools down faster because it has maximum temperature. Since, S α 1/dt [1+1]

22.

(i) It is based on mutual induction [1]

(i). It is step-up transformer because the number of turns in secondary coil is more than that in primary turn. [1]

(iii). Solution,

Given,

No. of turns in primary coil (Np)= 300 turns

Primary voltage (Vp) = 240V

No. of turns in secondary coil (Ns) = 500 turns

Secondary voltageor output voltage (Vs) = ?

We know that,

Np/Vp = Ns/Vs [1]

or, 300/240 = 500/Vs

or, Vs = 400 Volts [1]

23.

(i). The state of being healthy mentally, physically, socially and emotionally by balancing the time spent on online and offline activities is called digital well-being. [1]

(ii).

- It helps us manage screen time and avoid excessive use of mobile phones and gadgets. [1]
- It encourages us to take regular breaks and maintain good posture, reducing stress and improving focus. [1]

(Similar points can be considered).

(iii)

a. Beta b. Action Dash

[like others similar applications can be considered] [1]

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